EXPERIMENT - 64

1. Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, nums[]. The array contains exactly one element repeated once. Pass the array to a function named duplicate_element() using its address. Use pointers to pass the array between the calling function and duplicate_element(). The function duplicate_element() must:

- Find the only repeated element in nums[] and display the repeated element.
- Find the result in a single iteration through the array by using an efficient approach such as the XOR technique.
- Assume that every other element appears exactly once.

2. Algorithm

Main Program

Step 1: START
Step 2: Read n and declare the variable length array nums of size n.
Step 3: Read the n elements of nums.
Step 4: Call duplicate_element(&nums[0], n).
Step 5: STOP

Function duplicate_element()

Step 1: Set result = 0.
Step 2: Repeat for i = 0 to n − 1: result = result XOR nums[i] XOR i.
Step 3: Display result, which is the repeated element.

3. Test Case Table

#	Size	Array	Expected Output	Actual Output	Result
1	5	1 3 4 2 2	Repeated element: 2	Repeated element: 2	**PASS**
2	5	3 1 3 4 2	Repeated element: 3	Repeated element: 3	**PASS**
3	6	1 4 5 2 3 4	Repeated element: 4	Repeated element: 4	**PASS**
4	2	1 1	Repeated element: 1	Repeated element: 1	**PASS**
5	6	5 3 1 4 2 5	Repeated element: 5	Repeated element: 5	**PASS**

4. Source Code

dup64.c

```
/* Experiment 64: find the repeated element using XOR. */
#include <stdio.h>

void duplicate_element(int *nums, int n);

int main(void)
{
    int n, i;

    printf("Enter the size of the array: ");
    scanf("%d", &n);
    int nums[n];
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    duplicate_element(&nums[0], n);
    return 0;
}

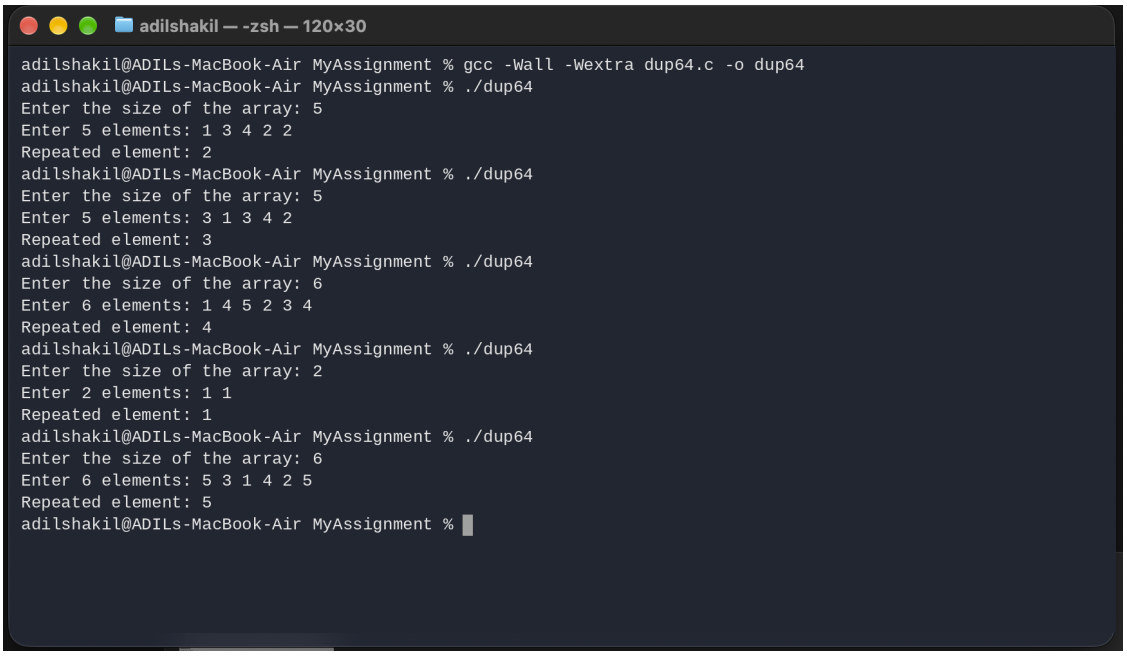
/* The array holds 1 to n-1 with one number repeated. XOR-ing every
   element with every index cancels each pair, so only the repeated
   number is left. One loop is enough. */
void duplicate_element(int *nums, int n)
{
    int i, result = 0;

    for (i = 0; i < n; i++)
        result ^= *(nums + i) ^ i;

    printf("Repeated element: %d\n", result);
}
```

5. Compilation and Output

Compiled with `gcc -Wall -Wextra dup64.c -o dup64` so the small warnings would show up too. It built clean.



All five passed, and the first three are the examples from the question.